Agile lab fat

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Question:Inventory Management System

problem statement:Ensure inventory levels are updated accurately after stock changes.

Tasks:

1.develop stock update logic

2.write junit test for stock updation scenarios

3.implement c1/cd pipeline with automated testing

setup version control using github

configure ci pipeline (eg .jenkins/githubb actions) to trigger on code commits

automate build process u maven

integrate automated unit tests in pipeline

configure cd pipeline for automatic deployment to staging/production

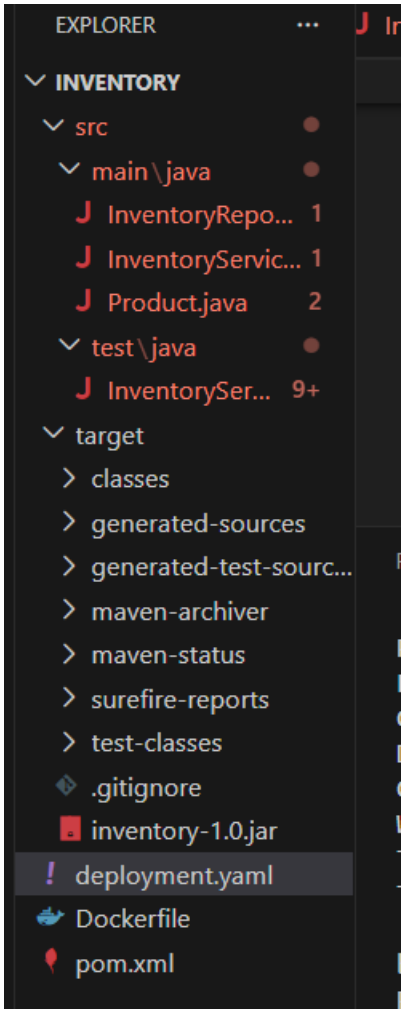
4.deploy services on docker and kubernetes

-contanerize application using docker(write dockerfile)

build and push docker image to container registry (docker h)

create kubernetes deployment and service yaml files

Code:



Product.java

java

```
public class Product {
    private int id;
    private String name;
    private int quantity;

    public Product(int id, String name, int quantity) {
        this.id = id;
        this.name = name;
        this.quantity = quantity;
```

```
    }

    public int getId() { return id; }
    public String getName() { return name; }
    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
    public void setName(String name) { this.name = name; }
}
```

InventoryRepository.java

java

```
import java.util.*;

public class InventoryRepository {
    private final Map<Integer, Product> store = new HashMap<>();

    public void save(Product product) {
        store.put(product.getId(), product);
    }

    public Optional<Product> findById(int id) {
        return Optional.ofNullable(store.get(id));
    }

    public List<Product> findAll() {
        return new ArrayList<>(store.values());
```

```
    }

    public boolean delete(int id) {
        return store.remove(id) != null;
    }

    public boolean exists(int id) {
        return store.containsKey(id);
    }
}
```

InventoryService.java

java

```
import java.util.*;

public class InventoryService {
    private final InventoryRepository repo;
    private int idCounter = 1;

    public InventoryService(InventoryRepository repo) {
        this.repo = repo;
    }

    // ADD new product to inventory
    public Product addProduct(String name, int quantity) {
        if (name == null || name.isBlank()) {
```

```
            throw new IllegalArgumentException("Product name cannot be empty");
        }
        if (quantity < 0) {
            throw new IllegalArgumentException("Quantity cannot be negative");
        }
        Product product = new Product(idCounter++, name, quantity);
        repo.save(product);
        return product;
    }

    // GET product
    public Product getProduct(int id) {
        return repo.findById(id)
            .orElseThrow(() -> new NoSuchElementException("Product not found: "
+ id));
    }

    // GET all products
    public List<Product> getAllProducts() {
        return repo.findAll();
    }

    // STOCK IN - increase stock
    public Product stockIn(int id, int quantity) {
        if (quantity <= 0) {
            throw new IllegalArgumentException("Stock-in quantity must be
positive");
```

```
        }
        Product product = getProduct(id);
        product.setQuantity(product.getQuantity() + quantity);
        repo.save(product);
        return product;
    }

    // STOCK OUT - decrease stock
    public Product stockOut(int id, int quantity) {
        if (quantity <= 0) {
            throw new IllegalArgumentException("Stock-out quantity must be
positive");
        }
        Product product = getProduct(id);
        if (product.getQuantity() < quantity) {
            throw new IllegalStateException("Insufficient stock for product: " + id);
        }
        product.setQuantity(product.getQuantity() - quantity);
        repo.save(product);
        return product;
    }

    // UPDATE stock directly
    public Product updateStock(int id, int newQuantity) {
        if (newQuantity < 0) {
            throw new IllegalArgumentException("Quantity cannot be negative");
        }
```

```
        Product product = getProduct(id);
        product.setQuantity(newQuantity);
        repo.save(product);
        return product;
    }

    // DELETE product
    public boolean deleteProduct(int id) {
        if (!repo.exists(id)) {
            throw new NoSuchElementException("Product not found: " + id);
        }
        return repo.delete(id);
    }
}
```

InventoryServiceTest.java

java

```
import org.junit.jupiter.api.*;
import java.util.NoSuchElementException;
import static org.junit.jupiter.api.Assertions.*;

class InventoryServiceTest {

    private InventoryService service;

    @BeforeEach
```

```
void setUp() {
    service = new InventoryService(new InventoryRepository());
}

// ── ADD PRODUCT ──────────────────────────────────────────

@Test
void addProduct_shouldCreateWithCorrectData() {
    Product p = service.addProduct("Rice", 100);
    assertEquals("Rice", p.getName());
    assertEquals(100, p.getQuantity());
    assertTrue(p.getId() > 0);
}

@Test
void addProduct_emptyName_shouldThrow() {
    assertThrows(IllegalArgumentException.class,
        () -> service.addProduct("", 100));
}

@Test
void addProduct_negativeQuantity_shouldThrow() {
    assertThrows(IllegalArgumentException.class,
        () -> service.addProduct("Rice", -10));
}
```

```
@Test
void addProduct_shouldAssignUniqueIds() {
    Product p1 = service.addProduct("Rice", 100);
    Product p2 = service.addProduct("Wheat", 200);
    assertNotEquals(p1.getId(), p2.getId());
}

// — STOCK IN ————————————————————————————————————

@Test
void stockIn_shouldIncreaseQuantity() {
    Product p = service.addProduct("Rice", 100);
    service.stockIn(p.getId(), 50);
    assertEquals(150, service.getProduct(p.getId()).getQuantity());
}

@Test
void stockIn_zeroQuantity_shouldThrow() {
    Product p = service.addProduct("Rice", 100);
    assertThrows(IllegalArgumentException.class,
        () -> service.stockIn(p.getId(), 0));
}

@Test
void stockIn_nonExistentProduct_shouldThrow() {
    assertThrows(NoSuchElementException.class,
```

```
        () -> service.stockIn(999, 50));
}

// — STOCK OUT ————————————————————————————————————

@Test
void stockOut_shouldDecreaseQuantity() {
    Product p = service.addProduct("Rice", 100);
    service.stockOut(p.getId(), 30);
    assertEquals(70, service.getProduct(p.getId()).getQuantity());
}

@Test
void stockOut_insufficientStock_shouldThrow() {
    Product p = service.addProduct("Rice", 20);
    assertThrows(IllegalStateException.class,
        () -> service.stockOut(p.getId(), 50));
}

@Test
void stockOut_exactQuantity_shouldResultInZero() {
    Product p = service.addProduct("Rice", 50);
    service.stockOut(p.getId(), 50);
    assertEquals(0, service.getProduct(p.getId()).getQuantity());
}
```

```
@Test
void stockOut_nonExistentProduct_shouldThrow() {
    assertThrows(NoSuchElementException.class,
        () -> service.stockOut(999, 10));
}

// ── UPDATE STOCK ──────────────────────────────────────────

@Test
void updateStock_shouldSetExactQuantity() {
    Product p = service.addProduct("Rice", 100);
    service.updateStock(p.getId(), 200);
    assertEquals(200, service.getProduct(p.getId()).getQuantity());
}

@Test
void updateStock_negativeQuantity_shouldThrow() {
    Product p = service.addProduct("Rice", 100);
    assertThrows(IllegalArgumentException.class,
        () -> service.updateStock(p.getId(), -5));
}

@Test
void updateStock_shouldNotAffectOtherProducts() {
    Product p1 = service.addProduct("Rice", 100);
    Product p2 = service.addProduct("Wheat", 200);
```

```
    service.updateStock(p1.getId(), 500);
    assertEquals(200, service.getProduct(p2.getId()).getQuantity());
}

// ── DELETE PRODUCT ──────────────────────────────────────

@Test
void deleteProduct_shouldRemoveFromInventory() {
    Product p = service.addProduct("Rice", 100);
    boolean result = service.deleteProduct(p.getId());
    assertTrue(result);
    assertThrows(NoSuchElementException.class,
        () -> service.getProduct(p.getId()));
}

@Test
void deleteProduct_nonExistentProduct_shouldThrow() {
    assertThrows(NoSuchElementException.class,
        () -> service.deleteProduct(999));
}

// ── DATA CONSISTENCY ────────────────────────────────────

@Test
void stockIn_multipleTimes_shouldAccumulate() {
    Product p = service.addProduct("Rice", 100);
```

```
        service.stockIn(p.getId(), 50);
        service.stockIn(p.getId(), 50);
        assertEquals(200, service.getProduct(p.getId()).getQuantity());
    }

    @Test
    void stockOut_shouldNotAffectOtherProducts() {
        Product p1 = service.addProduct("Rice", 100);
        Product p2 = service.addProduct("Wheat", 200);
        service.stockOut(p1.getId(), 50);
        assertEquals(200, service.getProduct(p2.getId()).getQuantity());
    }
}
```

---

pom.xml

xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.demo</groupId>
  <artifactId>inventory</artifactId>
  <version>1.0</version>
  <packaging>jar</packaging>
```

```
<properties>
  <maven.compiler.source>17</maven.compiler.source>
  <maven.compiler.target>17</maven.compiler.target>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  <junit.version>5.10.1</junit.version>
</properties>

<dependencies>
  <dependency>
    <groupId>org.junit.jupiter</groupId>
    <artifactId>junit-jupiter</artifactId>
    <version>${junit.version}</version>
    <scope>test</scope>
  </dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.2.5</version>
      <configuration>
        <useModulePath>false</useModulePath>
      </configuration>
    </plugin>
```

```
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-jar-plugin</artifactId>
      <version>3.4.1</version>
      <configuration>
        <archive>
          <manifest>
            <mainClass>InventoryService</mainClass>
          </manifest>
        </archive>
      </configuration>
    </plugin>
  </plugins>
 </build>
</project>
```

Dockerfile

dockerfile

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

---

deployment.yaml

yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: inventory-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: inventory
  template:
    metadata:
      labels:
        app: inventory
    spec:
      containers:
      - name: inventory
        image: abinaya160306/inventory:latest
        imagePullPolicy: Always
        ports:
        - containerPort: 8080
        resources:
          requests:
            memory: "256Mi"
            cpu: "250m"
          limits:
            memory: "512Mi"
```

```
          cpu: "500m"
        livenessProbe:
          tcpSocket:
            port: 8080
          initialDelaySeconds: 30
          periodSeconds: 10
        readinessProbe:
          tcpSocket:
            port: 8080
          initialDelaySeconds: 10
          periodSeconds: 5
---
apiVersion: v1
kind: Service
metadata:
  name: inventory-service
spec:
  type: NodePort
  selector:
    app: inventory
  ports:
  - protocol: TCP
    port: 80
    targetPort: 8080
    nodePort: 30008
```

Maven commands and git command:

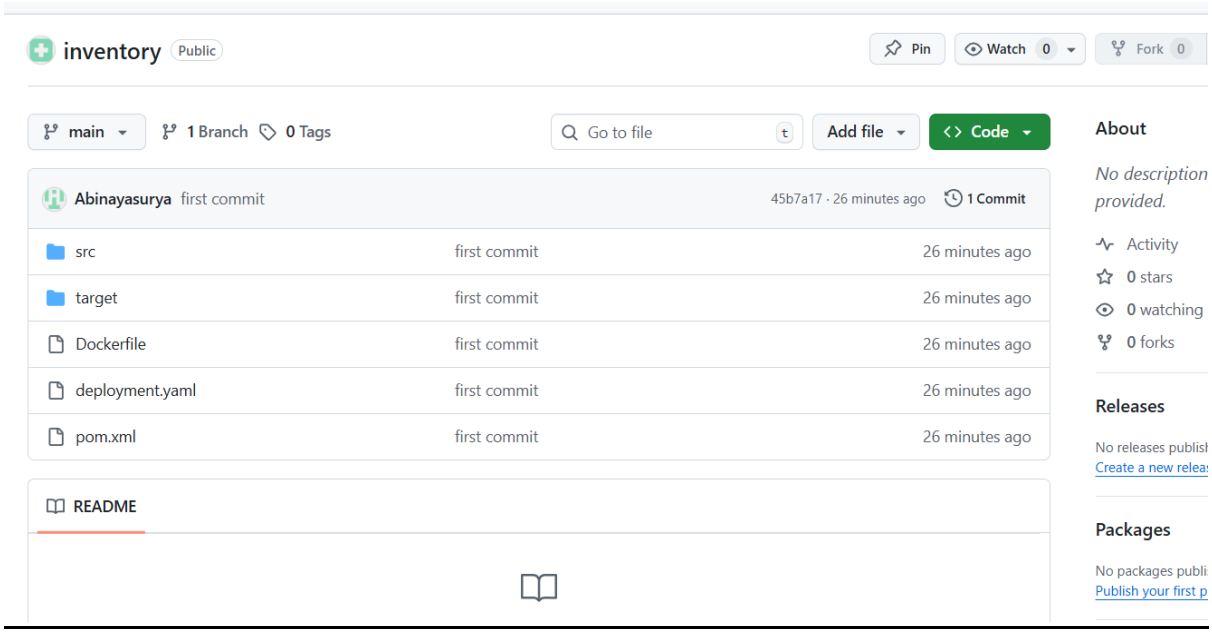
```
PS C:\Users\abina\Desktop\inventory> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:inventory >------------------------
[INFO] Building inventory 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ inventory ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ inventory ---
[INFO] skip non existing resourceDirectory C:\Users\abina\Desktop\inventory\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ inventory ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ inventory ---
[INFO] skip non existing resourceDirectory C:\Users\abina\Desktop\inventory\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ inventory ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ inventory ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running InventoryServiceTest
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.234 s -- in InventoryServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

```
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.160 s
[INFO] Finished at: 2026-04-13T10:21:57+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\abina\Desktop\inventory> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:inventory >------------------------
[INFO] Building inventory 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ inventory ---
[INFO] Deleting C:\Users\abina\Desktop\inventory\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ inventory ---
[INFO] skip non existing resourceDirectory C:\Users\abina\Desktop\inventory\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ inventory ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ inventory ---
[INFO] skip non existing resourceDirectory C:\Users\abina\Desktop\inventory\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ inventory ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ inventory ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
```

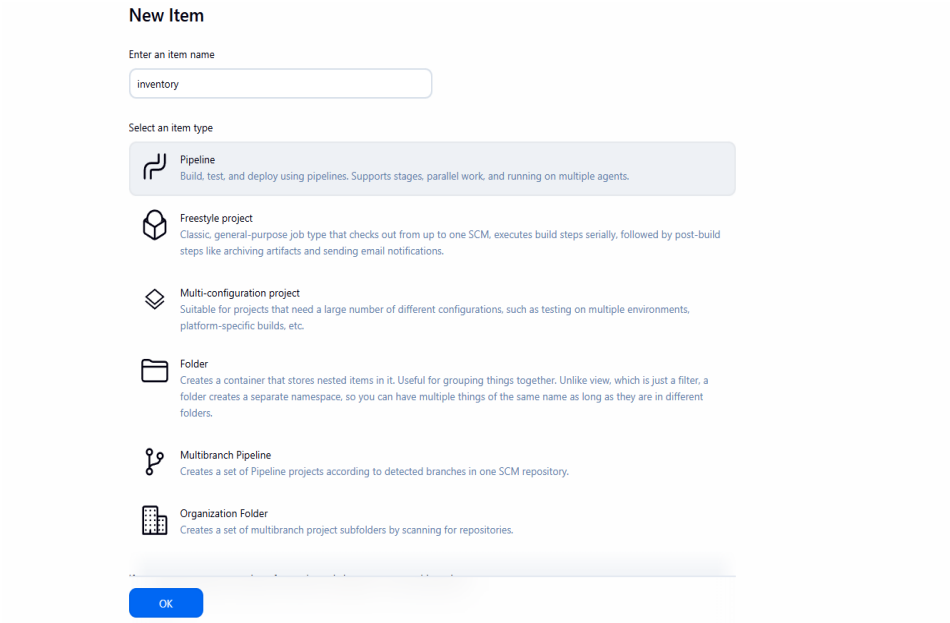
```
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ inventory ---
[INFO] Building jar: C:\Users\abina\Desktop\inventory\target\inventory-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  14.645 s
[INFO] Finished at: 2026-04-13T10:22:24+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\abina\Desktop\inventory> git init
Initialized empty Git repository in C:/Users/abina/Desktop/inventory/.git/
PS C:\Users\abina\Desktop\inventory> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next
 time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-InventoryServiceTest.xml', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\abina\Desktop\inventory> git add .
PS C:\Users\abina\Desktop\inventory> git commit -m "first commit"
[master (root-commit) 45b7a17] first commit
 20 files changed, 474 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 deployment.yaml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/InventoryRepository.java
 create mode 100644 src/main/java/InventoryService.java
 create mode 100644 src/main/java/Product.java
 create mode 100644 src/test/java/InventoryServiceTest.java
 create mode 100644 target/.gitignore
 create mode 100644 target/classes/InventoryRepository.class
 create mode 100644 target/classes/InventoryService.class
 create mode 100644 target/classes/Product.class
 create mode 100644 target/inventory-1.0.jar
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/InventoryServiceTest.txt
Indexing completed.   Java: Ready   Abinaya Srinivasan (now)   Ln 51, Col 20   Spaces: 2   UTF-8   CRLF   { } YAML   Go Live
```

```
 create mode 100644 target/inventory-1.0.jar
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/InventoryServiceTest.txt
 create mode 100644 target/surefire-reports/TEST-InventoryServiceTest.xml
 create mode 100644 target/test-classes/InventoryServiceTest.class
PS C:\Users\abina\Desktop\inventory> git branch -M main
PS C:\Users\abina\Desktop\inventory> git remote add origin https://github.com/Abinaya642-git/inventory.git
PS C:\Users\abina\Desktop\inventory> git push -u origin main
Enumerating objects: 38, done.
Counting objects: 100% (38/38), done.
Delta compression using up to 12 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (38/38), 16.15 KiB | 114.00 KiB/s, done.
Total 38 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Abinaya642-git/inventory.git
 * [new branch]      main -> main
```

Git output:



Jenkins pipeline:



Code:

```
pipeline {
  agent any

  tools {
    maven 'Maven3'
  }

  environment {
    DOCKER_IMAGE = "abinaya160306/inventory:latest"
  }

  stages {

    stage('Clone Repository') {
      steps {
        git branch: 'main',
          url: 'https://github.com/Abinaya642-git/inventory.git'
      }
```

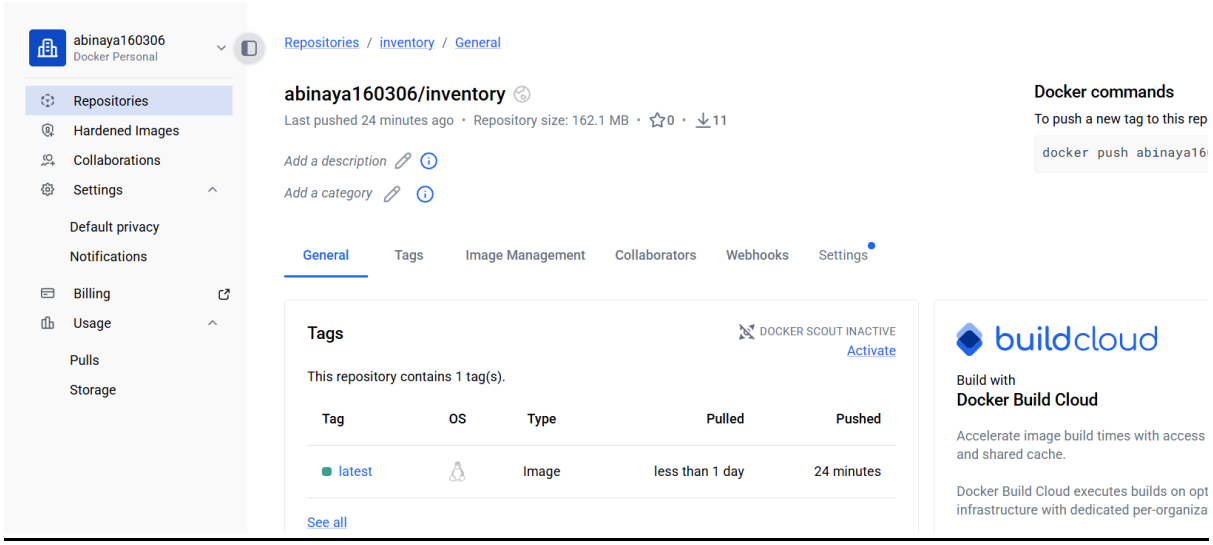
```
}

stage('Build & Run JUnit Tests') {
    steps {
        bat 'mvn clean test'
    }
    post {
        always {
            junit 'target/surefire-reports/*.xml'
        }
        failure {
            echo 'Tests FAILED — pipeline stopped. No Docker build.'
        }
    }
}

stage('Package JAR') {
    steps {
        bat 'mvn package -DskipTests'
    }
}

stage('Build Docker Image') {
    steps {
        bat "docker build -t %DOCKER_IMAGE% ."
    }
}

stage('Push to DockerHub') {
    steps {
        withCredentials([usernamePassword(
```

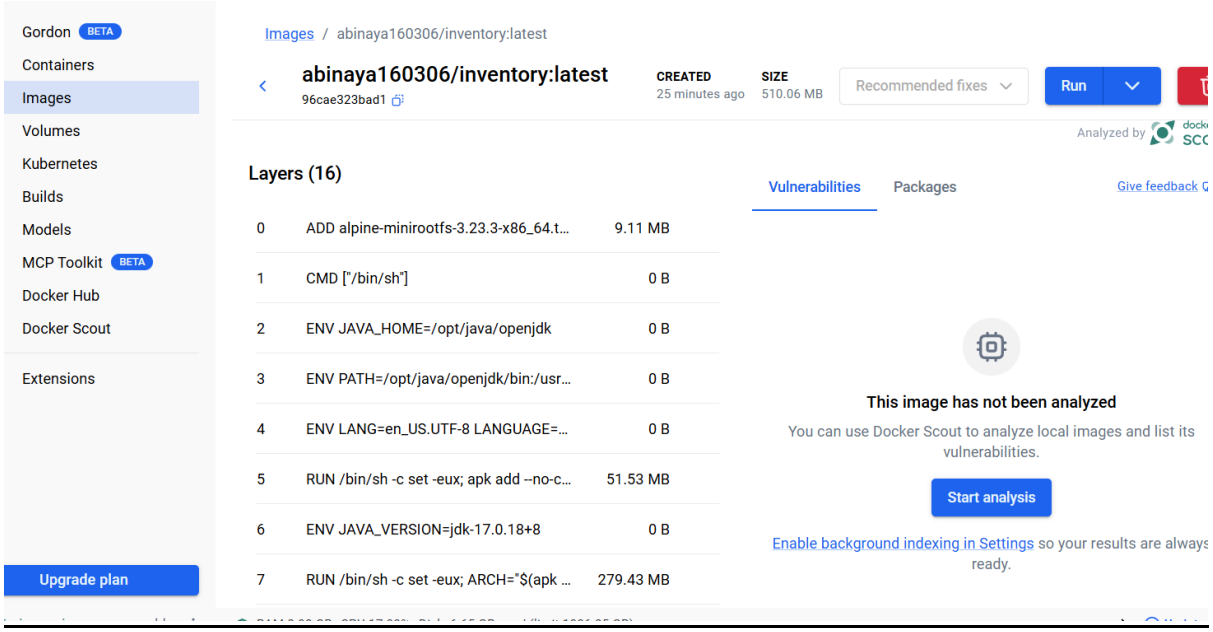
```
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )]) {
                bat """
                    docker login -u %DOCKER_USER% -p %DOCKER_PASS%
                    docker push %DOCKER_IMAGE%
                """
            }
        }
    }

    stage('Deploy to Kubernetes') {
        steps {
            withCredentials([file(credentialsId: 'kubeconfig', variable: 'KUBECONFIG')]) {
                bat 'kubectl apply -f deployment.yaml --validate=false'
            }
        }
    }
  }

  post {
    success {
        echo 'Pipeline complete. Inventory app deployed to Kubernetes.'
    }
    failure {
        echo 'Pipeline failed. Check test results.'
    }
  }
}
```

Output:

```
[Pipeline] {
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\inventory>kubectl apply -f deployment.yaml --validate=false
deployment.apps/inventory-deployment unchanged
service/inventory-service created
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline complete. Inventory app deployed to Kubernetes.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Dockerimage:


abinaya160306
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories / inventory / General
abinaya160306/inventory
Last pushed 24 minutes ago • Repository size: 162.1 MB • 0 • 11
Add a description
Add a category
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 1 tag(s).
Tag
OS
Type
Pulled
Pushed
latest
Image
less than 1 day
24 minutes
See all
Docker commands
To push a new tag to this rep
docker push abinaya16
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access and shared cache.
Docker Build Cloud executes builds on opt infrastructure with dedicated per-organiza


Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Images / abinaya160306/inventory:latest
abinaya160306/inventory:latest
96cae323bad1
CREATED
25 minutes ago
SIZE
510.06 MB
Recommended fixes
Run
Analyzed by
Layers (16)
0 ADD alpine-minirootfs-3.23.3-x86_64.t... 9.11 MB
1 CMD ["/bin/sh"] 0 B
2 ENV JAVA_HOME=/opt/java/openjdk 0 B
3 ENV PATH=/opt/java/openjdk/bin:/usr... 0 B
4 ENV LANG=en_US.UTF-8 LANGUAGE=... 0 B
5 RUN /bin/sh -c set -eux; apk add --no-c... 51.53 MB
6 ENV JAVA_VERSION=jdk-17.0.18+8 0 B
7 RUN /bin/sh -c set -eux; ARCH="$(apk ... 279.43 MB
Vulnerabilities
Packages
Give feedback
This image has not been analyzed
You can use Docker Scout to analyze local images and list its vulnerabilities.
Start analysis
Enable background indexing in Settings so your results are always ready.

Kubernets:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\abina> kubectl get svc
NAME                 TYPE         CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   80:30008/TCP   7m
inventory-service    NodePort    10.101.191.79    <none>        80:30009/TCP   7m
PS C:\Users\abina> kubectl get deployment
PS C:\Users\abina> kubectl get deployment
NAME                   READY    UP-TO-DATE     AVAILABLE     AGE
inventory-deployment   2/2      2              2             7m

PS C:\Users\abina> kubectl get pods
NAME                                    READY       STATUS     RESTARTS     AGE
inventory-deployment-5b99459db4-btrnh   1/1         Running    0            8m9s
inventory-deployment-5b99459db4-rwjnr   1/1         Running    0            8m9s
PS C:\Users\abina>
```